

LECTURE 1 PROBLEM SHEET

INTRODUCTION TO COUNTING ORBITS OF COREGULAR REPRESENTATIONS

Exercise 1 (Reinterpreting $H(\mathcal{O})$). Let K be a number field. Define its 3-Selmer group $\text{Sel}_3(K) := \{\delta \in K^\times : \delta \mathcal{O}_K = I^3 \text{ for some fractional ideal } I\} / (K^\times)^3$, i.e. the elements $\delta \in K^\times$ whose principal ideal is a cube, modulo cubes in $K^\times$.

- (a) When K is quadratic, give an isomorphism $H(\mathcal{O}_K) \xrightarrow{\sim} \text{Sel}_3(K)$.
- (b) If you are comfortable with (flat) cohomology, give an isomorphism $\text{Sel}_3(K) \xrightarrow{\sim} H^1(\mathcal{O}_K, \mu_3)$.

Exercise 2 (Verify the class-number formula). Let $\mathcal{O} = \mathbb{Z}[\sqrt{-2}]$, the ring of integers of $\mathbb{Q}(\sqrt{-2})$. Class-number tables give $\text{Cl}(\mathcal{O}) = \{1\}$, so $|\text{Cl}_3(\mathcal{O})| = 1$ and $|H(\mathcal{O})| = 1$ by a theorem from lecture. Verify the formula by enumerating admissible triples directly: show that the only admissible (I, δ) with $\mathcal{O} = \mathbb{Z}[\sqrt{-2}]$ is, up to equivalence, $(I, \delta) = (\mathcal{O}, 1)$.

Exercise 3 (Quadratic field count). Let $W_{\mathbb{R}} = \{f(x, y) = x^2 + axy + by^2 : a, b \in \mathbb{R}\} \cong \mathbb{R}^2$ be the space of “monic” binary quadratic forms. Let

$$G(\mathbb{R}) = \mathbb{G}_a(\mathbb{R}) \rtimes \mathbb{G}_m(\mathbb{R}) = \left\{ \begin{pmatrix} 1 & 0 \\ n & \lambda \end{pmatrix} : n \in \mathbb{R}, \lambda \in \mathbb{R}^\times \right\}$$

act on $W_{\mathbb{R}}$ via

$$(\gamma \cdot f)(x, y) = f((x, y) \cdot \gamma).$$

- (a) Show that, with coordinates (n, λ) on $G(\mathbb{R})$ and (a, b) on $W_{\mathbb{R}}$, the action is given by

$$(n, \lambda) \cdot (a, b) = (2n + a\lambda, n^2 + an\lambda + b\lambda^2).$$

- (b) Show that the discriminant $\text{Disc} = a^2 - 4b$ is invariant under the action of $G(\mathbb{Z})$. What about the action of $G(\mathbb{R})$?
- (c) Show that $G(\mathbb{Z})$ -orbits on $W_{\mathbb{Z}}$ are in bijection with isomorphism classes of rank-2 $\mathbb{Z}$ -algebras, via $(a, b) \mapsto \mathbb{Z}[x]/(x^2 + ax + b)$. Further show that discriminants match.
- (d) Show that $G(\mathbb{Z}) \backslash W_{\mathbb{Z}}$ has $\{0, 1\} \times \mathbb{Z}$ as a fundamental domain.
- (e) Show that the number of (integer) points with $0 < \text{Disc} < X$ (or $-X < \text{Disc} < 0$) is $X/2 + O(1)$.
- (f) Use this to deduce that the number of quadratic fields F with $0 < \text{Disc } F < X$ (or $-X < \text{Disc } F < 0$) is

$$\frac{1}{2\zeta(2)}X + o(X) = \frac{3}{\pi^2}X + o(X).$$

(Hint: the local density for maximality at a prime p is $1 - p^{-2}$).

- (g) Give an easier proof of the previous fact by counting fundamental discriminants.

Exercise 4 (reducible orbits of $\text{SL}_2 \curvearrowright \text{Sym}^3 \mathbb{Z}^2$). Recall the notion of an **admissible triple** $(\mathcal{O}, I, \delta)$:

- $\mathcal{O}$ is an oriented (i.e. equipped with an isomorphic $\pi: \mathcal{O}/\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}$) quadratic ring with $\text{Disc } \mathcal{O} \neq 0$;
- $I \subset \mathcal{O} \otimes \mathbb{Q}$ is a **fractional ideal**, i.e. an $\mathcal{O}$ -submodule which is $\mathbb{Z}$ -free of rank 2;
- $\delta \in (\mathcal{O} \otimes \mathbb{Q})^\times$ satisfies $I^3 \subset \delta \mathcal{O}$ and $N(I)^3 = N(\delta)$.

Recall also the group $H^{\text{red}}(\mathcal{O})$ of equivalence classes of those admissible triples (with first entry $\mathcal{O}$) such that δ is a cube. Let $\mathcal{I}_3(\mathcal{O})$ denote the group of 3-torsion fractional ideals (not ideal classes), i.e. fractional ideals I such that $I^3 = \mathcal{O}$.

- (a) Give an isomorphism $\mathcal{I}_3(\mathcal{O}) \xrightarrow{\sim} H^{\text{red}}(\mathcal{O})$.

- (b) Let $\mathcal{O} = \mathbb{Z}[\sqrt{-11}]$, a non-maximal quadratic order, and set $I := (2, (1 - \sqrt{-11})/2)$. Prove that $I \neq \mathcal{O}$, but $I \in \mathcal{I}_3(\mathcal{O})$.

Exercise 5 (The Delone–Faddeev correspondence). Let $V_{\mathbb{Z}} = \{Ax^3 + Bx^2y + Cxy^2 + Dy^3 : A, B, C, D \in \mathbb{Z}\}$, with the twisted $\mathrm{GL}_2(\mathbb{Z})$ -action $(\gamma \cdot f)(x, y) = \frac{1}{\det \gamma} f((x, y)\gamma)$. Prove that there is a natural bijection between $\mathrm{GL}_2(\mathbb{Z})$ -equivalence classes of elements of $V_{\mathbb{Z}}$ and isomorphism classes of cubic rings (rank-3 $\mathbb{Z}$ -algebras): given a cubic ring R with a normal basis $\langle 1, \omega, \theta \rangle$ (meaning $\omega\theta \in \mathbb{Z}$), show that the integers A, B, C, D defined by

$$\omega\theta = -AD, \quad \omega^2 = -AC - B\omega + A\theta, \quad \theta^2 = -BD - D\omega + C\theta$$

are the coefficients of a well-defined $\mathrm{GL}_2(\mathbb{Z})$ -equivalence class, and verify $\mathrm{Disc}(R) = \mathrm{Disc}(f)$ with $\mathrm{Disc}(f) = B^2C^2 - 4AC^3 - 4B^3D - 27A^2D^2 + 18ABCD$.

Exercise 6 (Classical cubic-field count). Combining Problem 5 with the averaging argument of Lecture 1 (adapted to $\mathrm{GL}_2(\mathbb{Z})$ on $V_{\mathbb{Z}}$), and using the following input from local densities:

- at each prime p , the density of $f \in V_{\mathbb{Z}_p}$ for which the associated cubic ring $R(f)$ is maximal at p is $\mu_p(\mathcal{U}_p) = (1 - p^{-2})(1 - p^{-3})$;
- a uniformity estimate: for each p , the number of f with $|\mathrm{Disc} f| < X$ and $R(f)$ not maximal at p is $O(X/p^2)$ uniformly in p ;

derive the Davenport–Heilbronn theorem on cubic fields:

$$N_3^+(X) = \frac{X}{12\zeta(3)} + o(X), \quad N_3^-(X) = \frac{X}{4\zeta(3)} + o(X).$$

(Hint: the raw $\mathrm{GL}_2(\mathbb{Z})$ -orbit count is $\pi^2 X / (12n_{\pm})$ with $n_+ = 6$, $n_- = 2$; the product of local densities is $\prod_p (1 - p^{-2})(1 - p^{-3}) = (\zeta(2)\zeta(3))^{-1}$, and $\zeta(2) = \pi^2/6$.)